

General information

How the instrument works

The Bot is active 24/7 for an end user, though the operating hours of the engine itself are only 18 hours a day. The engine is active daily from 06:06 to 00:06 (MSK, UTC+3). During this window, prices/volumes are streamed and metrics are recalculated as soon as new data arrives. From 00:06 to 06:06 the engine pauses market streams (a quiet window), so metrics are not updated until the next start. Again – the telegram part is still active, it is only bonds data which is stopped since the market is closed.

Real-time data sources. Market data (daily candle close and volume) is received through a broker market-data API. Reference bond parameters (nominal, accrued interest, maturity, coupons schedule, issue size, sector, country) are also pulled via API. Credit ratings and offer/call dates are obtained from Smart-Lab.

Refresh cycles. Prices/volumes update continuously during operating hours. Credit ratings and offer/call dates are refreshed every hour (outside the quiet window).

Instrument coverage and units

- **Universe / coverage.** The screener covers RUB-denominated bonds that match the following conditions: fixed coupon (floating-coupon bonds are excluded), no amortization, not perpetual. Instruments without a credit rating on Smart-Lab are excluded from the universe.
- **Price field.** Price is shown as an estimated money price per 1 bond in RUB: $\text{Price} = (\text{close_quote} / 100) * \text{Nominal}$. This is a practical approximation derived from the daily candle close quote.
- **Accrued interest (ACI).** Accrued Interest is taken from the API and is in RUB per 1 bond.
- **Yields.** YTM/YTC/YTW are annualized (per year) and displayed in %. Internally they are stored as decimals (e.g., 0.125 = 12.5%).
- **Durations.** Macaulay/Modified durations are shown in years.
- **DtA (days-to-event).** DtM/DtC/DtW are calendar days.
- **Volume.** Vol is daily trading volume in lots or in rubles from the candle stream.
- **Issue size.** Issue Size is the number of bonds in the issue (as provided by the API). Issue (RUB) equals $\text{IssueSize} * \text{Nominal}$.

Cashflow conventions and settlement rule

All cashflow-based metrics are computed on actual coupon dates (no month-end approximations). Time to each cashflow is measured on an ACT/365 basis:

$$t_i = (\text{Date}_i - \text{Today}) / 365$$

Dirty price (initial investment / PV) is computed in RUB per 1 bond:

$$\text{PV} = \text{Price} + \text{AccruedInterest}$$

Settlement adjustment (important). When you buy today, settlement is T+1 (adjusted for non-trading days). If the first upcoming coupon's fix date is earlier than the settlement date, that coupon is excluded from the cashflows (because the buyer is not entitled to it).

Yield metrics (YTM, YTC, YTW)

All yields are computed as XIRR on actual dates. In simplified discounted-cashflow form, YTM is the annual rate y that solves:

$$0 = -PV + \sum_{i=1..N} [CF_i / (1 + y)^{t_i}] + Nominal / (1 + y)^{t_M}$$

Yield to Call/Offer (YTC) is computed in the same way, but only coupons up to the offer/call date are included, and the nominal redemption is assumed to occur on the offer/call date at nominal (par):

$$0 = -PV + \sum_{i=1..K} [CF_i / (1 + y)^{t_i}] + Nominal / (1 + y)^{t_C}$$

Yield to Worst (YTW) selects the lower of YTM and YTC if an offer/call date exists; otherwise YTW equals YTM:

If offer/call exists: $YTW = \min(YTM, YTC)$; else: $YTW = YTM$

Practical notes: yields do not include broker commissions, taxes, FX conversion, or execution slippage. Future cashflows are taken from the provided coupon schedule; bonds with floating coupons are excluded from the universe.

DtA metrics (DtM, DtC, DtW)

Days-to-event metrics are calendar days:

$$DtM = (MaturityDate - Today).days$$

$$DtC = (OfferDate - Today).days \quad (\text{if no offer/call: } DtC = 0)$$

DtW (days to worst) is chosen based on which yield is worse. For non-weighted values it follows YTM vs YTC; for weighted values it follows $YTM(w)$ vs $YTC(w)$:

If $YTC < YTM$ then $DtW = DtC$, else $DtW = DtM$. If $YTC(w) < YTM(w)$ then $DtW(w) = DtC$, else $DtW(w) = DtM$.

Duration and convexity (Macaulay, Modified)

For a chosen yield y (YTM, YTC, or YTW) and cashflows CF_i at times t_i :

$$PV(CF_i) = CF_i / (1 + y)^{t_i}$$

$$MacaulayDuration = [\sum_i (t_i * PV(CF_i))] / PV$$

$$ModifiedDuration = MacaulayDuration / (1 + y)$$

$$Convexity = [\sum_i (t_i * (t_i + 1) * CF_i / (1 + y)^{t_i + 2})] / PV$$

DPP (price sensitivity to a 1 p.p. yield move)

DPP estimates the relative price change ($\Delta P / P$) using the duration-convexity approximation with $\Delta Y = 0.01$:

$$DPP_down_1pp = +ModifiedDuration * \Delta Y + 0.5 * Convexity * (\Delta Y^2)$$

$$DPP_up_1pp = -ModifiedDuration * \Delta Y + 0.5 * Convexity * (\Delta Y^2)$$

Weighted “(w)” metrics (credit-risk weighting)

Fields with the “(w)” suffix apply a simplified credit-risk adjustment using the issuer’s Smart-Lab credit rating. The idea: future cashflows are multiplied by an estimated survival probability $S(t)$. This produces ‘survival-weighted’ cashflows and, from them, weighted yields and weighted risk metrics.

Step 1 — default probabilities by rating and hazard rates.

For each rating bucket we use cumulative default probabilities PD_1Y, PD_2Y, PD_3Y and convert them into piecewise hazard rates:

$$S1 = 1 - PD_1Y; \quad S2 = 1 - PD_2Y; \quad S3 = 1 - PD_3Y$$

$$\lambda_{1} = -\ln(S1)$$

$$\lambda_{2} = \ln(S1 / S2)$$

$$\lambda_{3} = \ln(S2 / S3)$$

Step 2 — survival probability to any date.

$$t = (\text{TargetDate} - \text{Today}).\text{days} / 365$$

$S(t)$ is then computed piecewise:

$$\text{if } t \leq 1: \quad S(t) = \exp(-\lambda_1 * t)$$

$$\text{if } 1 < t \leq 2: \quad S(t) = \exp(-\lambda_1 - \lambda_2 * (t - 1))$$

$$\text{if } t > 2: \quad S(t) = \exp(-\lambda_1 - \lambda_2 - \lambda_3 * (t - 2))$$

Step 3 — build survival-weighted cashflows.

$$\text{Coupon}_i(w) = \text{Coupon}_i * S(t_i)$$

$$\text{Nominal}(w) = \text{Nominal} * S(t_{\text{redemption}})$$

The initial purchase cashflow (-PV) is not weighted.

Step 4 — compute “(w)” metrics.

YTM(w), YTC(w), and YTW(w) are computed as XIRR using the weighted cashflows. Weighted DtW(w) is chosen by comparing YTC(w) vs YTM(w) (the smaller yield determines the ‘worst’ date).

Duration(w), Convexity(w), and DPP(w) are computed on the survival-weighted cashflows but using the corresponding non-weighted yield (e.g., Duration_YTM(w) uses weighted cashflows with $y = \text{YTM}$; Duration_YTC(w) uses weighted cashflows with $y = \text{YTC}$). This keeps sensitivity measures comparable to the base yield used in discounting.

Ratings update every hour, so all “(w)” metrics can change after a rating refresh even if the market price stays the same.

Liquidity metrics (Volume, Volume/Issue size)

VolMoney is an approximate traded value proxy based on the latest price:

$$\text{VolMoney} = \text{Vol} * \text{Price}$$

Volume / Issue size (%) is computed as:

$$\text{VolIssuePct} = (\text{Vol} * 100) / \text{IssueSize}$$

Assumptions and limitations

- Near real-time does not mean tick-perfect: upstream API latency, network issues, and Telegram rate limits can cause delays.
- All calculations exclude broker commissions, taxes, and any execution slippage. Use the screener for analytics, not as a trade confirmation – in other words **you are the only one responsible for the actions performed, bot does not provide any calls to actions.**
- Offer/call modelling assumes redemption at nominal (par) on the offer/call date. If the real terms differ (premium/partial/complex), results may deviate.
- The “(w)” model is a simplified survival-weighting approach. It is not a full structural credit model and does not model recovery rates or spread term structure.
- If a bond temporarily has no valid last price, it can be removed from the active calculation universe until prices become available again.

Screener function (/screener, /sc)

1) What “Screener” is and what it does

Screener is a real-time, interactive bond shortlist: it pulls the latest available data, instantly recalculates results when you change settings, and lets you explore the universe of bonds without leaving the chat. In a few taps you can scan, filter, sort, and group instruments to quickly find candidates that match your yield, risk, liquidity, and maturity preferences. See a list of bonds with a configurable set of columns (“Fields”).

- **Sort** the list by different metrics (yield, duration, etc.) and choose **ascending/descending** order.
- **Group** the list (e.g., by sector or rating) and show the “top N” bonds inside each group.
- **Filter** bonds by numeric ranges (e.g., YTW between 10–15) and by categorical values (e.g., ratings, sectors).
- Optionally view **Watchlist-only** results and/or exclude bonds via a **Blacklist**.

The top of the screener message always includes:

- Current **Sort**, **Order**, **Group**, **Group size**, and a detailed **Filters** summary.

Default behavior on first launch

- Default fields: Name, **YTW**, Price.
 - Default rating filter: only AAA, AA, A, BBB. (Bonds without a rating will not pass that filter.)
 - Default sort: YTW, descending (**the sorting parameter is always shown in Bold**).
 - Default grouping: Sector.
-

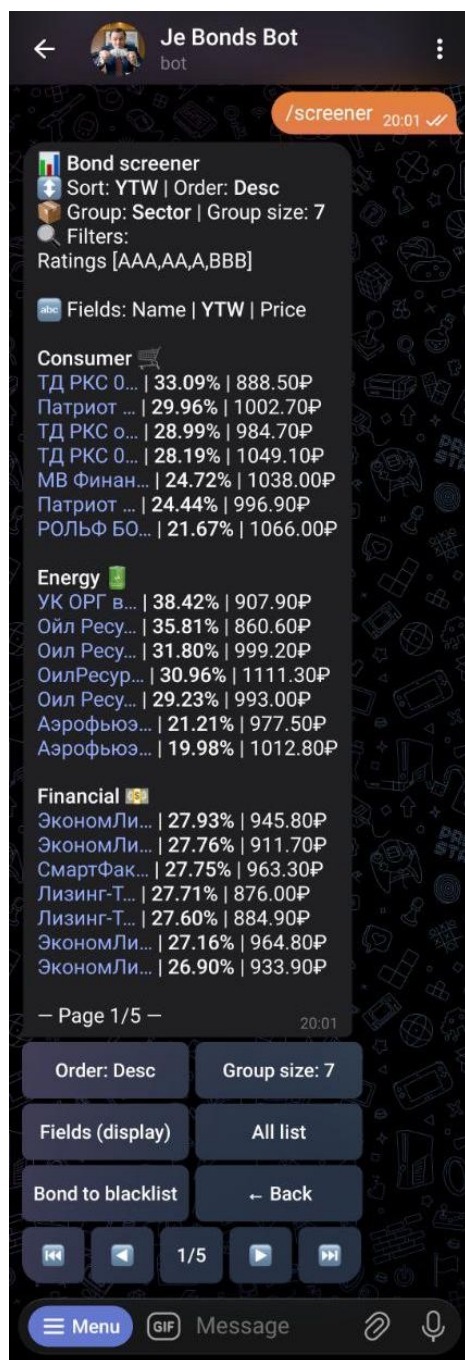
2) How to open the Screener

Users start it via the bot commands:

- /screener (or /sc) to open the screener message and interactive keyboard.

Important UI note

- Screener is a single message that updates (edits) as the user presses buttons.
- If the user runs /screener again, the bot deactivates/removes the keyboard on the previous screener message and creates a new instance.



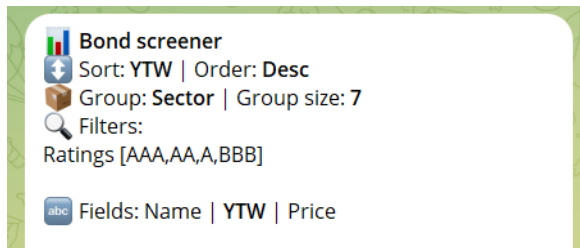
3) Understanding the Screener screen layout

3.1 Header (top of the message)

Header shows:

- Title: “ **Bond screener**”
- Sort: **Sort:** | **Order:** Asc/Desc
- Group: **Group:** | **Group size:**

- Filters: a multi-line summary of what's currently active (yields, duration, price, ratings, sectors, countries, coupons, offer mode, etc.).
- Fields: shows the currently active fields displayed



3.2 Rows (bond results)

Each bond is displayed as one line with selected fields. Typically the **Name** (or **Ticker**, depending on field order) is a clickable link to the bond page (Tinkoff link).

3.3 Pagination footer

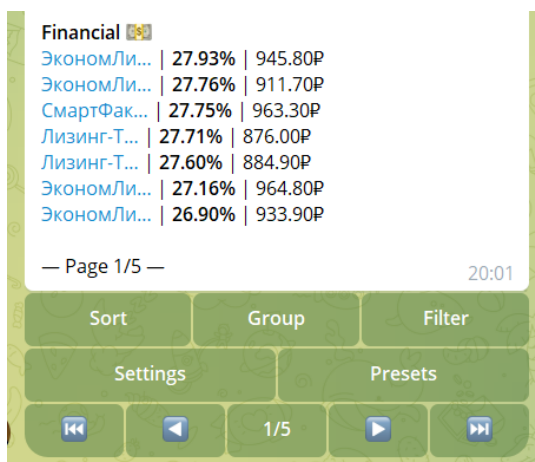
At the very bottom of the text content, the screener appends:

- — Page X/Y —

3.4 Navigation row (page controls)

Below the buttons, the screener shows navigation controls:

- ☐ first page
- ◀ previous page
- X/Y (not clickable; informational)
- ▶ next page
- ☐ last page



4) Main menu (what each main button does)

When the screener is in the **main menu**, the keyboard shows these buttons:

- Sort
- Group
- Filter

- **Fields**
- **Settings**
- **Presets**

The bot can also return you to the main menu automatically:

- If you stay inside a submenu and don't press anything for **30 seconds**, it returns to the main menu keyboard.

5) Sort menu (Sort button)

5.1 Sort menu overview

Press **Sort** to open the sorting menu. Buttons include:

- **Yield ▶**
- **Duration ▶**
- **% change if yield ↓ 1% ▶**
- **DtA ▶**
- **Last price**
- **Volume**
- **Issue size**
- **Volume/Issue size**
- **← Back**

What happens when you choose a sort

- The chosen metric becomes the active sort field.
- The list rerenders immediately with the new ordering.
- The screener ensures the active sort field is included in visible fields in bold (so users can see what they're sorting by).



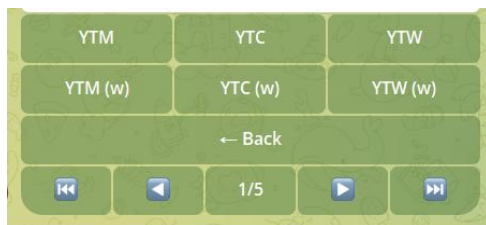
5.2 Sort ▶ Yield

Inside **Yield ▶**, users can sort by:

- YTM, YTC, YTW
- Weighted versions: YTM (w), YTC (w), YTW (w)

What “(w)” means in UI

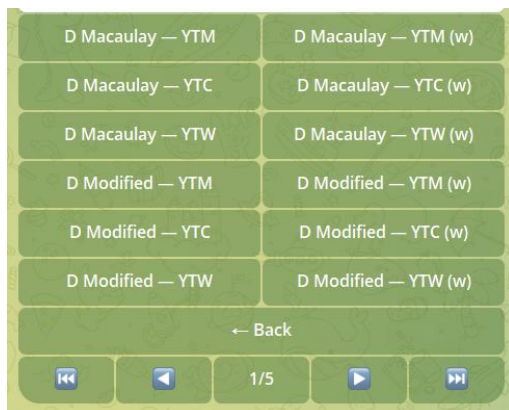
- In this bot, “(w)” is treated as a separate precomputed metric (weighted variant) and can be sorted/filtered independently.



5.3 Sort ▶ Duration

Inside **Duration** ▶, users can sort by:

- Macaulay duration for YTM / YTC / YTW
- Modified duration for YTM / YTC / YTW
- And weighted versions (w) for each variant



5.4 Sort ▶ % change if yield ↓ 1%

Inside **% change if yield ↓ 1%** ▶, users can sort by:

- % change if yield ↓ 1% — YTM and (w)
- % change if yield ↓ 1% — YTC and (w)
- % change if yield ↓ 1% — YTW and (w)

These are the “DPP” scenario metrics (price change sensitivity scenario).



5.5 Sort ▶ DtA

Inside **DtA** ▶, users can sort by:

- DtM (days to maturity)
- DtC (days to next call/offer)
- DtW (days to “worst”/effective term)



5.6 Sort single-click items

The following items in Sort menu apply immediately without a submenu:

- **Price**
- **Volume**
- **Issue size**
- **Volume/Issue size**
- **Reset sort** (returns to default sort settings)

6) Group menu (Group button)

Press **Group** to open grouping options. Choices:

- **No group**
- **Credit rating**
- **Sector**
- **Country**
- **Coupons per year**

- Put/Call option
- Time to maturity (buckets)
- Issue size (buckets)
- ← Back



The chosen type of grouping is ticked.

6.1 What grouping does

- The screener splits results into groups (e.g., each sector).
- For each group, it shows **a header with group name and count**, then shows only the **top N items** inside that group, where N = **Group size**.

6.2 Group size (how many items per group)

- “Group size” is configured in **Settings** (see section 9).
- Example: if Group size = 7, the screener shows up to 7 bonds per group.

6.3 How to remove grouping

- Select **No group**.

Bond screener

Sort: YTW | Order: Desc

Group: Time to maturity (buckets) | Group size: 7

Filters:

Ratings [AAA,AA,A,BBB]

Fields: Name | YTW | Price

<=3m

ИК Айген...	23.19%	1000.00P
Глоракс ...	19.94%	997.00P
ЕвроТран...	19.00%	997.20P
Сэтл Гру...	18.00%	992.00P
Ренессанс...	17.55%	990.20P
Кировски...	17.34%	994.90P
Атомстро...	17.18%	984.80P

3-6m

СмартФак...	27.75%	963.30P
ЭкономЛи...	27.16%	964.80P
Делимоби...	26.75%	1002.80P
ПР-Лизин...	18.97%	1005.50P
Ред Софт...	17.56%	989.10P
Трансмаш...	17.04%	975.70P
Трансмаш...	15.94%	971.60P

6m-1y

УК ОРГ в...	38.42%	907.90P
ЭкономЛи...	27.93%	945.80P
ЭкономЛи...	26.90%	933.90P
СмартФак...	26.63%	955.00P
АйДи Кол...	24.74%	963.00P
Дарс-Дев...	22.04%	1037.40P
ВОКСИС-04	21.08%	1003.60P

— Page 1/3 —

20:01

- Submenus: **Yield** ▸, **Duration** ▸, **% change if yield ↓ 1%** ▸, **DtA** ▸
- Range filters: **Price**, **Volume (P)**, **Par value**, **Issue size**
- Categorical filters: **Credit ratings**, **Sectors**, **Countries**, **Coupons per year**

- Mode filter: **Put/Call option** (toggle cycles)
- **Clear filters**
- ← **Back**



8) How range filters work (the most important part)

Whenever you tap a numeric filter (like **Price** or **YTW**), the bot will ask you to **type a range condition**.

8.1 Accepted input formats

Range parsing supports:

- min-max (inclusive)
- $\geq x$, $\leq y$
- $> x$, $< y$
- Decimal separator: . or , is accepted (comma is converted to dot).

8.2 Percent sign behavior (critical)

If the user types a trailing %, the bot interprets the numbers as percent and **divides by 100** internally. Example:

- Input 10-15% becomes 0.10–0.15 internally.

Important note

- For yield filters in this screener, users should typically enter **numbers without %** (e.g., 10-15) because yields are displayed in percent in the UI.
- Only include % if your data model truly expects fractional values for that particular metric.

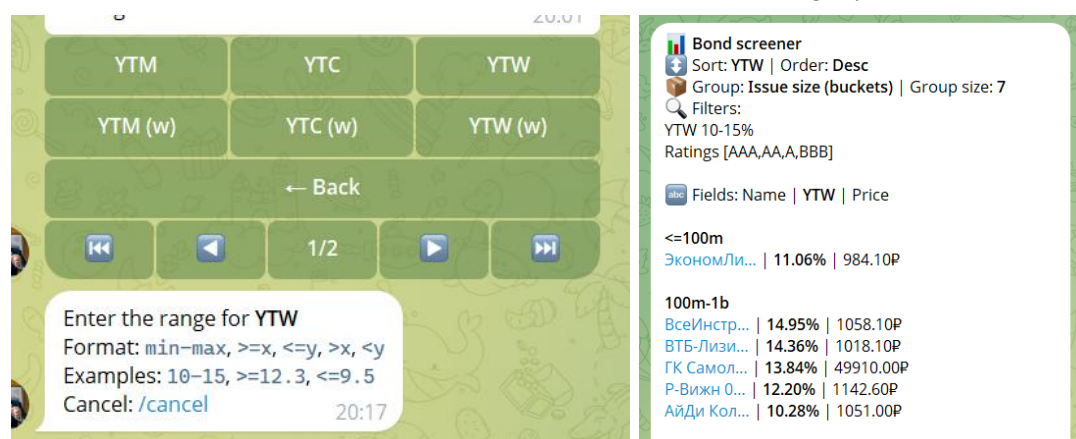
8.3 What happens if the input is invalid

If the bot can't parse the range, it replies with an error message and examples like:

- 10-15, >=12.3, <=9.5, >12.3, <9.5
Or with % variants for percent units.

8.4 After a valid input

- The filter is stored.
- The screener rerenders immediately.
- The bot deletes the prompt and the user's numeric message (to keep chat clean).

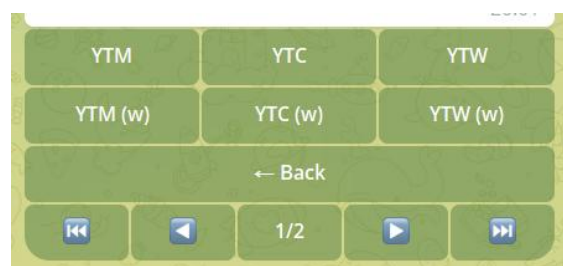


9) Filter submenus (Yield / Duration / % change / DtA)

9.1 Filter ► Yield

Buttons:

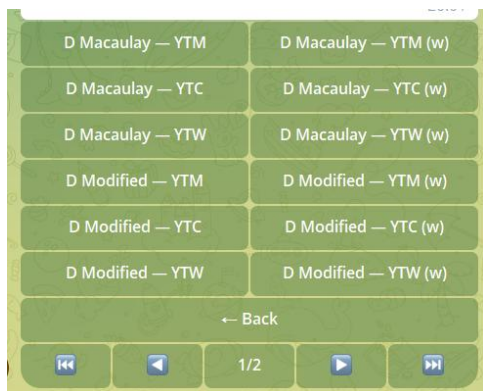
- YTM, YTC, YTW, and weighted versions (w)
- ← Back



9.2 Filter ► Duration

Buttons include:

- Macaulay and Modified duration variants for YTM/YTC/YTW, plus weighted variants
- ← Back



9.3 Filter ▶ % change if yield ↓ 1%

Buttons include:

- % dP@-1% — YTM, YTC, YTW and weighted variants
- ← Back

9.4 Filter ▶ DtA

Buttons include:

- DtM, DtC, DtW
- ← Back

10) Categorical filters (Ratings / Sectors / Countries / Coupons)

10.1 Credit ratings

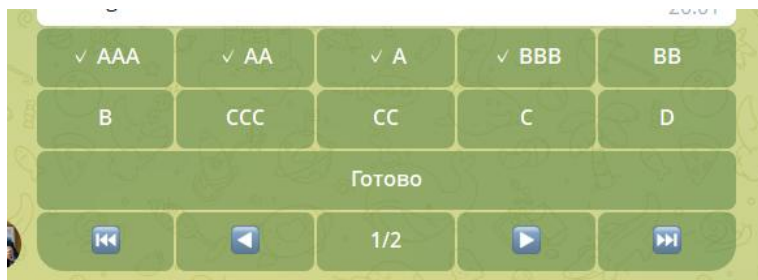
Press **Credit ratings** to open a picker where each rating can be toggled on/off. Selected items show ✓.

Buttons:

- Rating buttons (AAA, AA, ... down to D) in a fixed order.
- Готово ("Done") returns to Filter menu.

Default

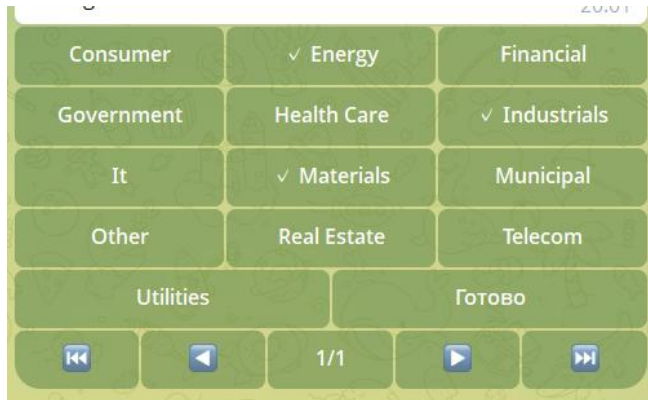
- By default, ratings filter is set to AAA, AA, A, BBB.



10.2 Sectors

Press **Sectors** to open a sector list:

- Each sector can be toggled; selected sectors show ✓.
- Unknown/empty sector is displayed as **Other**.
- **Готово** returns to Filter menu.



10.3 Countries

Press **Countries** to open the country list:

- Empty/None country is shown as **Unknown**.
- Toggle selection with ✓.
- **Готово** returns to Filter menu.



10.4 Coupons per year

Press **Coupons per year** to open a picker with values:

- 0, 1, 2, 4, 6, 12, 24 (each toggles, ✓ indicates selected)
- **Готово** returns to Filter menu.

11) Put/Call option filter (cycle button)

In the Filter main menu, **Put/Call option** cycles through modes:

- **including**: show bonds both with and without an offer date (no restriction)
- **excluding**: show only bonds **without** put/call offer
- **only**: show only bonds **with** put/call offer

12) Clear filters

Press **Clear filters** to reset the filter state to an **empty FilterState** (i.e., it clears everything).

14) Settings menu (Settings button)

Settings buttons include:

- **Order: Desc/Asc** (toggle)
- **Group size: N** (set numeric)
- **Fields (display)** (opens Fields)
- **Watchlist / All list** (toggle)
- **← Back**

14.1 Order

Toggles sort order:

- Desc ↔ Asc

14.2 Group size

Prompts user:

- “Введите новое значение Group size (≥ 1):”
User types an integer. This controls how many items show per group.

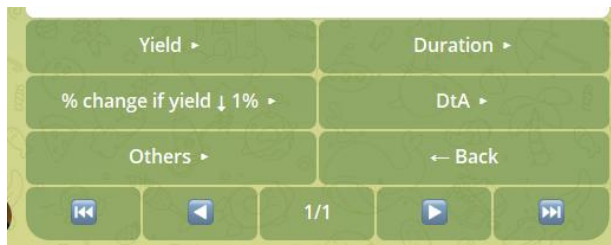
14.3 Fields (display)

Fields control which columns are shown in each bond line.

Fields root screen

The Fields menu is structured by categories:

- **Yield**
- **Duration**
- **Dpp**
- **DtA**
- **Others**
- **Back**

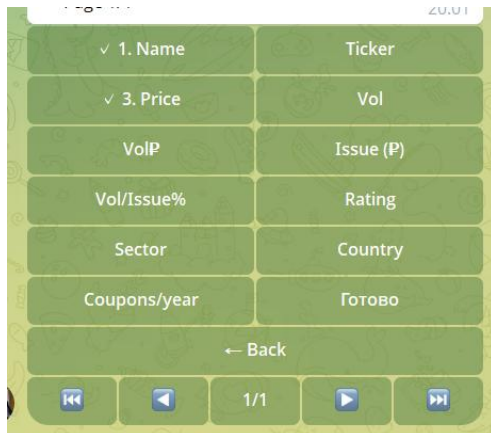


Inside a category (toggling fields)

In each category screen:

- Tap a field to toggle it on/off.
- Selected fields show a ✓.

- Press ← **Back** to return to Fields root.



Safety rules

- If the user removes both Name and Ticker, the bot automatically adds Ticker back so that each row still has an identifier.
- The bot also ensures the current **sort field** is always included in displayed fields.

14.4 Watchlist / All list

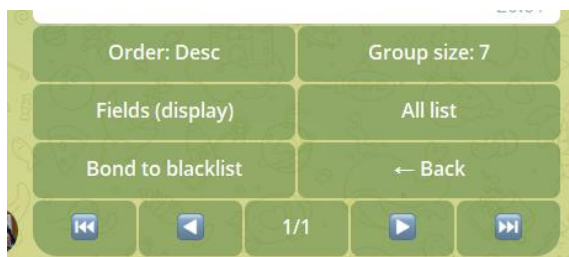
Toggles watchlist mode:

- In **Watchlist** mode, screener shows only tickers in the user watchlist (if the watchlist is empty, the filter effectively doesn't limit results).

14.5 Bond to blacklist

Prompts user:

- “Введите тикеры для чёрного списка через пробел/запятую.”
User enters tickers separated by spaces or commas, which get added to blacklist.

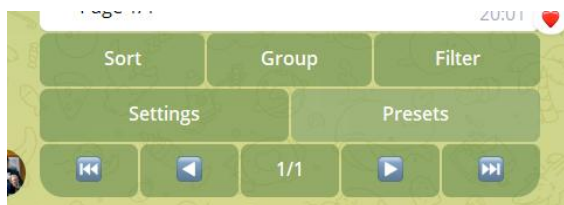


15) Presets menu (Presets button)

15.1 What presets do

A preset applies a full configuration:

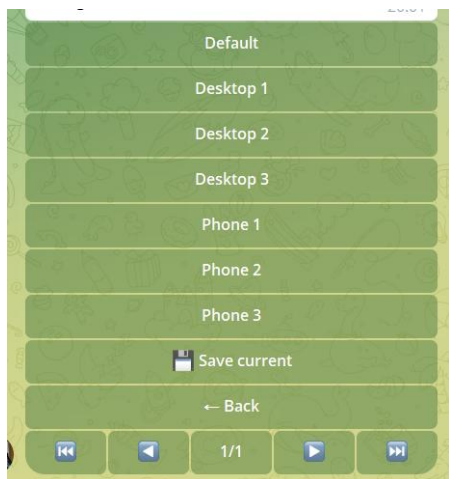
- **Fields**
- **Sort field**
- Whether sort is weighted (w)
- **Order** (asc/desc)
- **Group mode**



15.2 Built-in presets (what each contains)

Built-in presets include:

- Default
- Desktop 1/2/3
- Phone 1/2/3



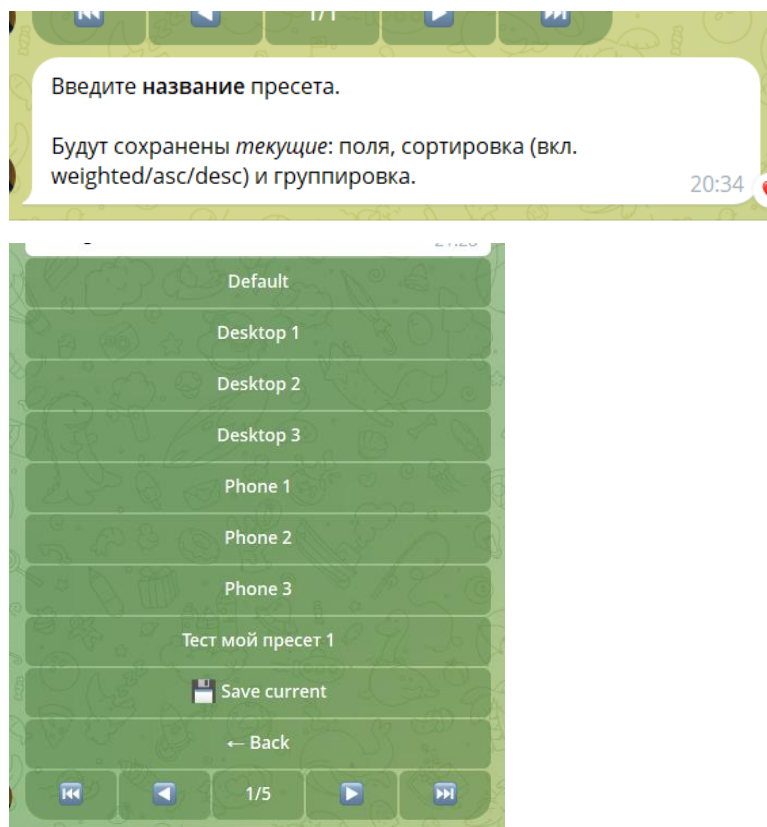
Their exact configurations are:

- **Default:** fields [Name, YTW, Price], sort YTW, desc, group sector
- **Desktop 1:** [Name, YTW, Price, Rating, DtW, D_mac@YTW, VolP, Vol/Issue%], group sector
- **Desktop 2:** [Name, YTW, YTW (w), Price, VolP, Issue (P), Vol/Issue%, DtW], group sector
- **Desktop 3:** [Name, YTW, Sector, D_mac@YTW, % change if yield ↓ 1% @YTW, VolP, Coupons/year], group rating
- **Phone 1:** [Name, YTW, Price, Rating], group sector
- **Phone 2:** [Name, YTW, Price, DtW], group sector
- **Phone 3:** [Name, YTW, Price, VolP], group rating

15.3 Saving user presets

The presets menu also includes:

-  **Save current** — saves the current configuration as a user preset (the bot will ask for a name).
- User-saved presets appear as additional buttons.
- ← **Back** returns to main.



16) Quick “How do I...?” recipes

“I want only bonds with YTW between 12 and 16”

1. Open **Filter**.
2. Tap **Yield** ►.
3. Tap **YTW**.
4. Reply with 12-16. (No % sign.)

“I want only bonds with put/call offer”

1. Open **Filter**.
2. Tap **Put/Call** until it shows **only**.

“I want to see fewer items per group”

1. Open **Settings**.
 2. Tap **Group size: N**.
 3. Enter a smaller integer (e.g., 5).
-

Alerts function (/alerts, /alert, /al)

1) What Alerts Do

Alerts are a rule-based monitoring and notification system for bonds. Instead of you repeatedly checking the market manually, an alert acts like an automated watcher: it continuously (or on a defined schedule) evaluates the latest bond data against a set of criteria you define, and notifies you when a bond enters the “match” state—meaning the bond now satisfies your rule.

An alert is made of:

1. **Scope** — which bonds the alert should track (whole market, your watchlist, or specific bonds).
2. **Filters** — narrow the universe (sector, country, issue size, number of coupons per year, par value, Put/Call, or by removing specific bonds)
3. **Conditions** — numeric or categorical checks (e.g., “Yield > 12%”, “Duration < 2y”, “Rating is BBB or higher” etc.). Conditions are combined via a logical expression (AND/OR + parentheses).
4. **Trigger mode** — how you want to be notified:
 - **Once** (on crossing into “true”)
 - **Perpetually**
 - **Daily digest**
 - **Weekly digest**

In short: **Alerts turn your bond criteria into an automated detector.** When the market changes (in real time) and a bond crosses into your defined criteria, you don’t have to discover it—the bot pushes it to you, in the format and frequency you chose (immediate alerts or scheduled digests).

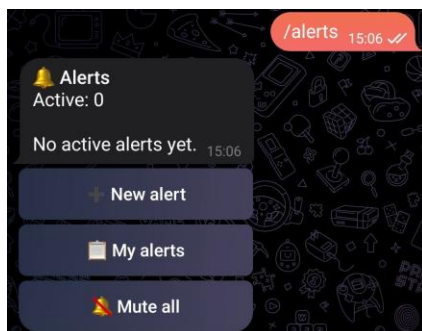
2) How to Open Alerts

Commands

You can open the Alerts UI using any of these commands in the bot chat:

- /alerts
- /alert
- /al

After running the command, the bot shows the **Alerts Home screen**.



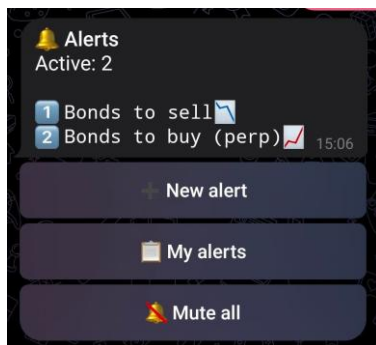
3) Alerts Home Screen (Root)

The home screen is the landing page where you see how many alerts you have and how to proceed.

3.1 What You See

The message contains:

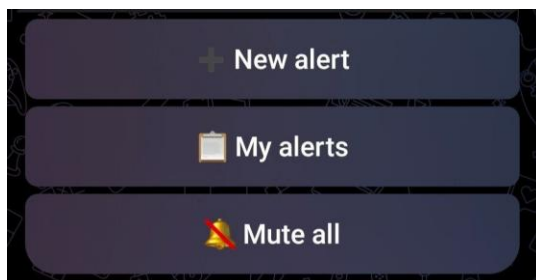
- Title: “🔔 **Alerts**”
- Count of active (enabled) alerts: “**Active: N**”
- List of up to 20 active alerts, numbered with emoji digits (e.g., 🅑...)
- If you have no active alerts: “**No active alerts yet.**”
- If notifications are muted: a line “🔕 **Muted (notifications are paused).**”



3.2 Buttons on Home Screen

Home screen has exactly 3 buttons (each on its own row):

1. **+ New alert**
 - Starts the creation wizard.
2. **📄 My alerts**
 - Opens the Manage screen.
3. **🔕 Mute all**
 - Opens the Mute menu where you snooze notifications.



4) “📄 My alerts” — Manage Screen

The Manage screen is where you view, select, edit, pause/resume, delete, or open matches for an alert.

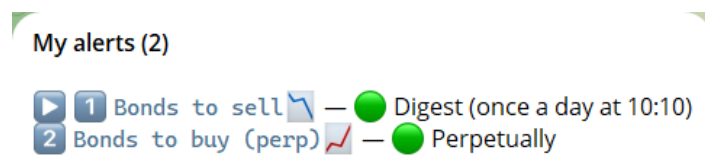
4.1 What You See

At the top:

- “**My alerts (N)**” where N is total number of alerts.

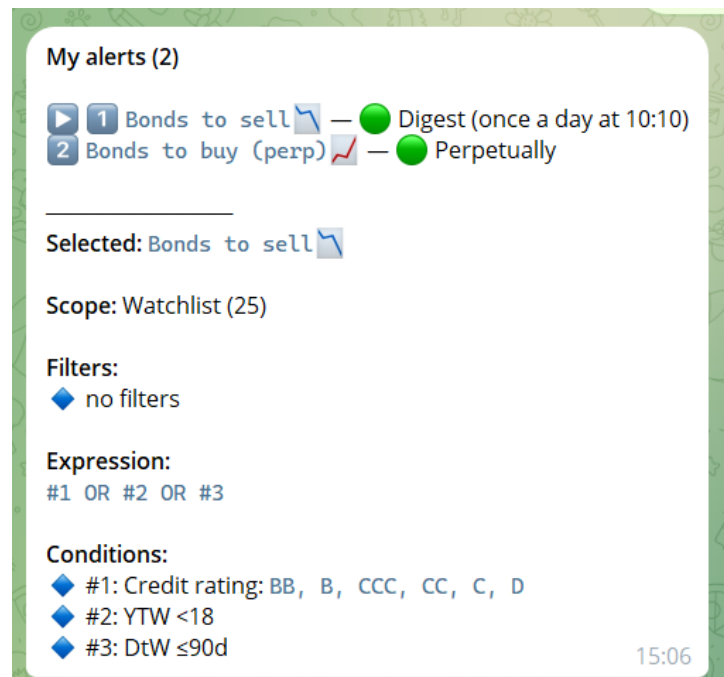
Then a numbered list of **all alerts** (enabled and paused):

- The **selected** alert is prefixed with “▶”
- Each line shows:
 - alert number emoji (e.g., 2)
 - alert label
 - status icon: ● if enabled, ⏸ if paused
 - mode title: e.g., “Once”, “Perpetually”, “Daily digest ...”, “Weekly digest ...”



Below the list there is a **Selected** section that displays the selected alert’s configuration in detail:

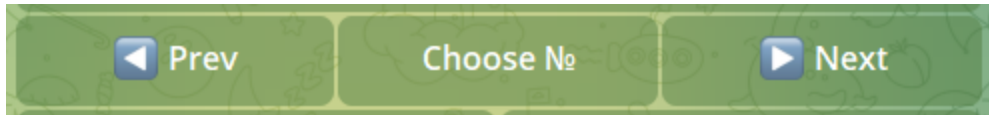
- **Scope** (what is being tracked)
- **Filters** (each active filter shown as a bullet line)
- **Expression** (if the alert uses the expression builder)
- **Conditions** list (each condition numbered, with ranges/values)



4.2 Selecting an Alert

There are two ways:

- **Tap an alert line** in the list (each line is a button) to select it.
- Or press **Choose №** to open a numeric picker (useful if you have many alerts).



if “Choose №”:



4.3 Buttons in Manage Screen (Global Controls)

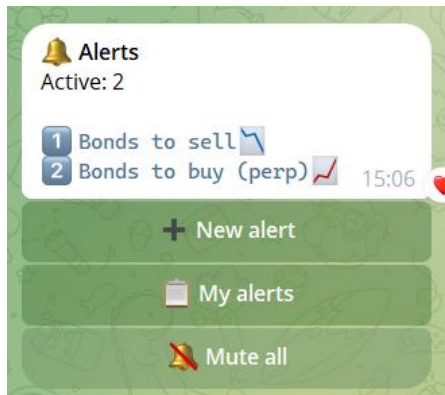
The Manage screen keyboard contains these controls:

1.  **Edit**
 - Opens the wizard pre-filled with the selected alert's settings.
2.  **Delete**
 - Deletes the selected alert immediately.
3.  **Pause / ► Turn on**
 - Toggles the selected alert ON/OFF (enabled ↔ paused).
4.  **Prev**
 - Selects the previous alert in the list.
5. **Choose №**
 - Opens numeric selection screen.
6.  **Next**
 - Selects the next alert in the list.
7.  **Back**
 - Returns to Alerts Home.
8.  **New alert**
 - Starts a new alert wizard (without losing the current alert list).



5) “🔕 Mute all” — Snooze Notifications Menu

Muting does not delete alerts; it temporarily stops sending alert notifications.

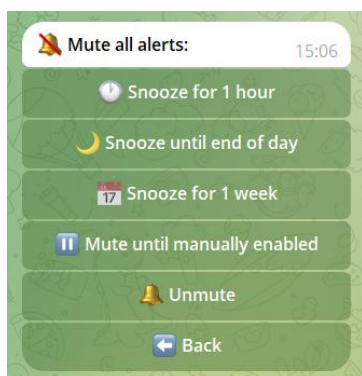


5.1 What You See

The screen title: “**Mute all alerts:**” and buttons described below.

5.2 Buttons

1. 🕒 **Snooze for 1 hour**
2. 🌙 **Snooze until end of day**
3. 📅 **Snooze for 1 week**
4. ☐ **Mute until manually enabled**
5. 🔔 **Unmute**
6. ← **Back** (returns to Alerts Home)



Time zone note: the bot uses **MSK (Europe/Moscow)** for certain “end of day” and digest scheduling calculations.

6) Creating / Editing an Alert — The 5-Step Wizard

Press **+** **New alert** or **✎** **Edit** to open the wizard. □

The wizard has 5 steps:

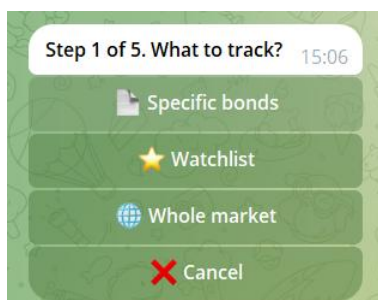
1. **Scope**
2. **Filters**
3. **Conditions & Expression**
4. **Trigger mode**
5. **Name**
→ then **Summary** → **Save**

Step 1 of 5 — Scope (What bonds to track)

Screen Purpose

Choose the bond universe:

-  **Specific bonds** (manual tickers list)
-  **Watchlist** (your saved watchlist)
-  **Whole market** (all bonds available)
-  **Cancel** (exit wizard, go back)



If you pick “ **Specific bonds**”

The bot switches to an input screen where you must send tickers (one message) as text.

- Separate tickers by spaces or commas.
- The bot validates them and stores as the scope list.
- After successful entry, you continue to Step 2.

Step 1 of 5. Securities

Enter tickers/ISIN/UID as a list (separated by spaces or new lines).
Example:
RU000A105KD1 RU000A105LQ3

15:06

Back Cancel

Step 2 of 5 — Filters (Narrow the universe)

Screen Purpose

Filters constrain the scope before evaluating conditions.

The Step 2 screen displays:

- Title: **Step 2 of 5**
- Current **Scope** with number of currently applicable for the scope and applied filters bonds (**dynamic**)
- A list of currently active filters (or “(none)”)
- Buttons to set filters or proceed.

Step 2 of 5. Add filters?

Scope: Watchlist (25)

Current filters:
◆ no filters

15:06

Continue

Issue size Sectors ▶

Countries ▶ Coupon per year ▶

Par value

Put/Call option (toggle) Remove bonds

Clear Back

Cancel

Buttons in Step 2

1. → **Continue (Moves to step 3)**

2. **Issue size**
 3. **Sectors**
 4. **Countries**
 5. **Coupon per year**
 6. **Par value**
 7. **Put/Call option**
 8. **Remove bonds**
 9. 🗑️ **Clear** (clears all filters)
 10. ⬅️ **Back** (returns to Step 1)
 11. ❌ **Cancel** (exits wizard)
-

2.1 Range Filters (Issue size, Par value)

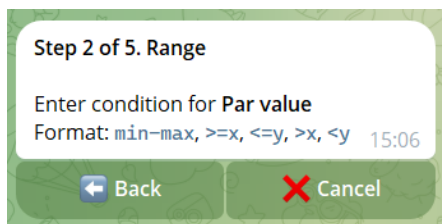
When you press **Issue size** or **Par value**, the bot asks you to type a numeric range.

Accepted formats:

- min-max (e.g., 100-500)
- >=min and >min (e.g., >=100 or >100)
- <=max and <max (e.g., <=500 or <500)

Buttons on the range input screen:

- ⬅️ **Back** (returns to Step 2)
- ❌ **Cancel** (exits wizard)

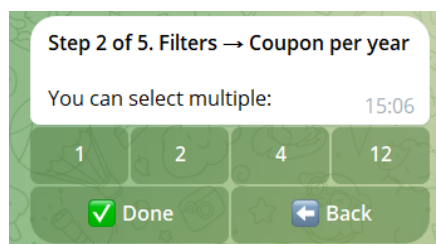
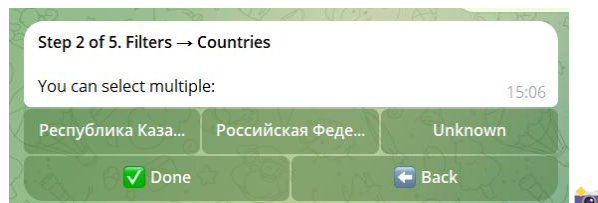
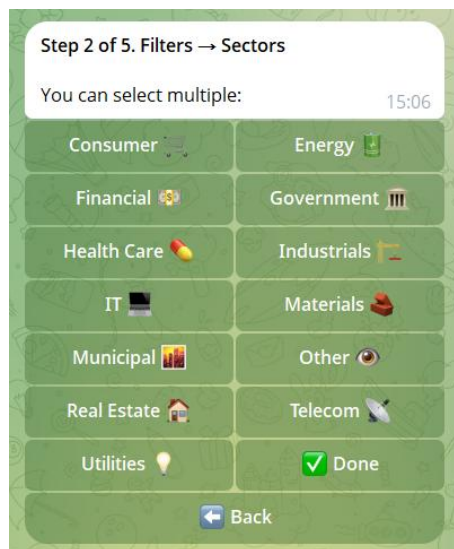


2.2 List Filters (Sectors, Countries, Coupon per year)

When you press **Sectors**, **Countries**, or **Coupon per year**, you get a checklist menu.

How it works:

- Each item toggles on/off.
- Selected items show a “✅”.
- Press **Done** to confirm and return to Step 2.
- Press ⬅️ **Back** to return to Step 2 without changing further.



2.3 Put/Call Option (Toggle)

Put/Call option is a toggle filter with 3 states:

1. **Any** (no filter)
2. **Exclude Put/Call bonds**
3. **Only Put/Call bonds**

Users press the button repeatedly to cycle through these states.

Current filters:

◆ no filters

Current filters:

◆ Put/Call option: excluded

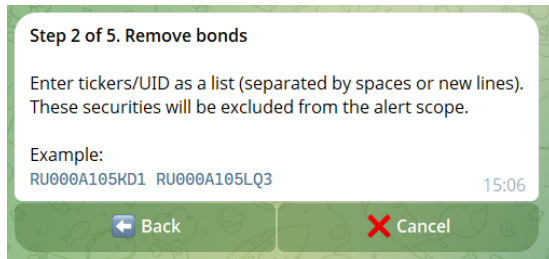
Current filters:

◆ Put/Call option: only

2.4 Remove bonds (Exclude tickers)

This lets the user provide a list of tickers that will be excluded even if they match.

The bot prompts the user to send tickers, similar to “Specific bonds”.



Step 3 of 5 — Conditions & Logical Expression

This is the most important step: define what “matching” means.

Screen Purpose

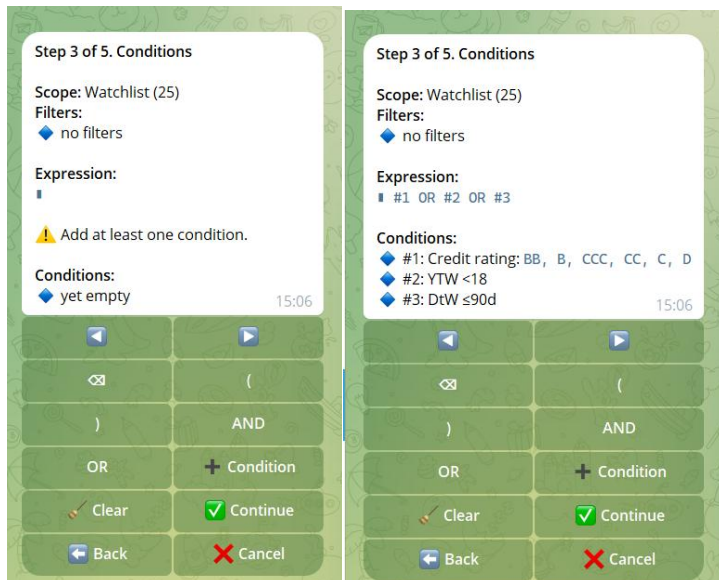
The Step 3 screen shows:

- Current expression (with a cursor “|”)
- Conditions list (numbered)
- A keyboard to build logic and add/delete conditions.

The expression preview looks like:

C1 AND (C2 OR C3)

(with a cursor marker where insertion happens).



Buttons in Step 3 (Expression Keyboard)

Cursor movement

- ← Move cursor left
- → Move cursor right

Insert logic tokens

- (insert opening parenthesis
-) insert closing parenthesis
- AND
- OR

Condition controls

- **+ Condition** — add a new condition (opens parameter selection)
- **Delete** — delete token left of cursor
- **Clear** — clears the expression and all conditions

Navigation

- **Continue** — validates expression and goes to Step 4
- **Back** — returns to Step 2
- **Cancel** — exits wizard

Step 3 of 5. Conditions

Scope: Whole market (255)

Filters:

- ◆ Sectors: Consumer, Energy, Government, Health Care, Industrials, IT, Materials, Municipal, Other, Telecom, Utilities
- ◆ Put/Call option: excluded
- ◆ Remove bonds: Айди Коллект выпуск 3, Артген выпуск 1, Бизнес Альянс 1P-01, Борец Капитал 1P-01, Дарс-Девелопмент 001P-02, Делимобиль 001P-02, Заслон выпуск 1, ЛСР выпуск 8, МаксимаТелеком выпуск 1, МВ ФИНАНС 001P-04...

Expression:

#1 AND ((#2 AND #3) || OR (#4 AND #5))

Conditions:

- ◆ #1: DtW 120-300d
- ◆ #2: YTW >20
- ◆ #3: Credit rating: AAA, AA, A, BBB
- ◆ #4: YTW >18
- ◆ #5: Credit rating: AAA, AA, A

15:06

Navigation buttons: Back, Clear, + Condition, Continue, Cancel, AND, OR, (,), Delete.

Expression Validation Rules (What “**Continue**” checks)

When you press **Continue**, the bot validates that:

- Parentheses are balanced
 - Token order is logically valid
 - Conditions referenced exist
- If validation fails, it shows an error (e.g., “Invalid expression: ...”).

Adding a Condition (**+ Condition flow**)

Press **+ Condition** → you enter a 2-stage selector:

1. **Choose a parameter group** (Yield, Duration, Ratings, etc.)
2. **Choose a specific parameter** within that group
Then enter a value/range or pick from a list (ratings).

Condition: select a parameter 15:06

Yield ▾	Duration ▾
Change in yield ▾	DtA ▾
Credit ratings	Last price
Volume	Volume / Issue size
← Back	✖ Cancel

Select a parameter: 15:06

YTM	YTC
YTW	YTM (w)
YTC (w)	YTW (w)
← Back	✖ Cancel

Select a parameter: 15:06

D Macaulay @YTM	D Macaulay @YTC
D Macaulay @YTW	D Modified @YTM
D Modified @YTC	D Modified @YTW
D Macaulay @YTM (w)	D Macaulay @YTC (w)
D Macaulay @YTW (w)	D Modified @YTM (w)
D Modified @YTC (w)	D Modified @YTW (w)
← Back	✖ Cancel

Select a parameter: 15:06

% change if yield ↓ 1% — YTM	% change if yield ↓ 1% — YTC
% change if yield ↓ 1% — YTW	% change if yield ↓ 1% — YTM ...
% change if yield ↓ 1% — YTC (...)	% change if yield ↓ 1% — YTW ...
← Back	✖ Cancel

Numeric conditions (range input)

For most numeric parameters you'll be asked to type a range using the same formats as filters (min-max, >=, <=, >, <).

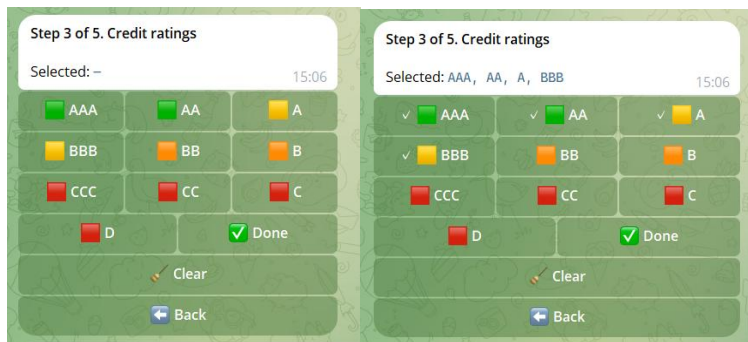
Step 3 of 5. Range

Set a condition for % change if yield ↓ 1% — YTM
Format: min-max, >=x, <=y, >x, <y 15:06

← Back ✖ Cancel

Credit rating condition (list picker) – the only categorical value which serves as a condition since credit ratings apart from sector, issue size, number of coupons per year etc. is dynamic and can be changed, thus should be duly monitored.

If you choose a rating parameter, you get a list of ratings you can toggle and save.



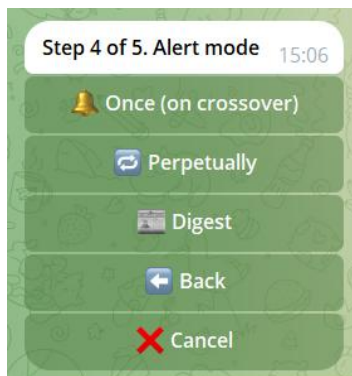
Step 4 of 5 — Trigger Mode (When to notify)

Screen Purpose

Choose how often you want notifications.

Buttons:

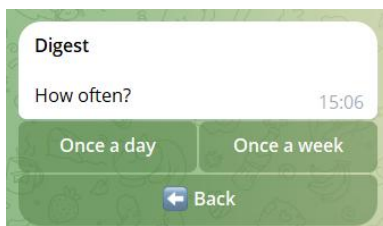
- **Once** (notify only when a bond becomes matching instantly)
- **Perpetually** (repeat notifications while matching, subject to cooldown of 5 minutes to mitigate potential spam)
- **Digest** (send one digest per day or per week)
- **← Back**
- **● Cancel**



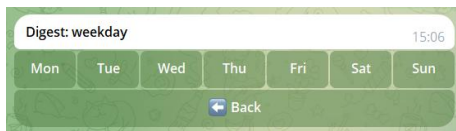
Digest scheduling (Daily/Weekly)

If you select a digest trigger, you then configure:

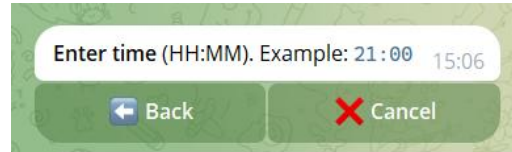
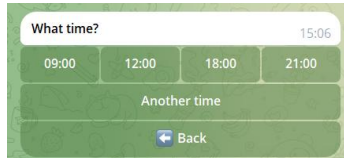
- Daily vs Weekly (kind)



- For weekly: select weekdays



- Select time presets (09:00, 12:00, 18:00, 21:00) or enter custom HH:MM



- The time input is validated strictly as HH:MM format.

Step 5 of 5 — Name Your Alert

Screen Purpose

Give the alert a friendly label.

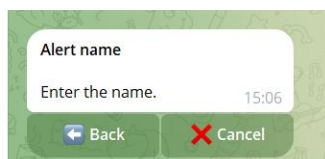
Buttons:

-  **Enter name** (switches to text input)
-  **Use default name** (bot generates a name automatically)
-  **Back**
-  **Cancel**



When entering a name:

- You type the name as a normal chat message.
- The bot saves it and proceeds to Summary.



Summary & Save

After Step 5, the bot shows a **Summary** page including:

- Name

- Scope
- Filters
- Expression / Conditions
- Trigger mode

Buttons:

-  **Save**
-  **Back**
-  **Cancel**

Review & confirm

Alert name: Bonds to buy (perp)

Scope: Whole market (255)

Filters:

- ◆ Sectors: Consumer , Energy , Government , Health Care , Industrials , IT , Materials , Municipal , Other , Telecom , Utilities
- ◆ Put/Call option: excluded
- ◆ Remove bonds: Айди Коллект выпуск 3, Артген выпуск 1, Бизнес Альянс 1P-01, Борец Капитал 1P-01, Дарс-Девелопмент 001P-02, Делимобиль 001P-02, Заслон выпуск 1, ЛСР выпуск 8, МаксимаТелеком выпуск 1, МВ ФИНАНС 001P-04...

Expression:

```
#1 AND ( ( #2 AND #3 ) OR ( #4 AND #5 ) )
```

Conditions:

- ◆ #1: DtW 120-300d
- ◆ #2: YTW >20
- ◆ #3: Credit rating: AAA, AA, A, BBB
- ◆ #4: YTW >18
- ◆ #5: Credit rating: AAA, AA, A

Trigger: Perpetually
15:06

Save

Back

Cancel

After saving:

- The alert is stored and becomes active (unless paused).
- The bot may perform an **initial check** and show bonds that match right now (paginated view).

Currently matches the condition (initial check)

Bonds to buy (perp)

Snapshot: 15:44 09.01.2026 MSK

- ◆ УК ОРГ выпуск 3 — DtW 120-300d (fact 225d); YTW >20 (fact 38.42%); Credit rating: AAA, AA, A, BBB (fact BBB-)
- ◆ Делимобиль 001P-05 — DtW 120-300d (fact 120d); YTW >20 (fact 26.79%); Credit rating: AAA, AA, A, BBB (fact BBB+)
- ◆ БОКСИС-04 — DtW 120-300d (fact 289d); YTW >20 (fact 21.23%); Credit rating: AAA, AA, A, BBB (fact BBB+)

— Page 1/1 —
15:44

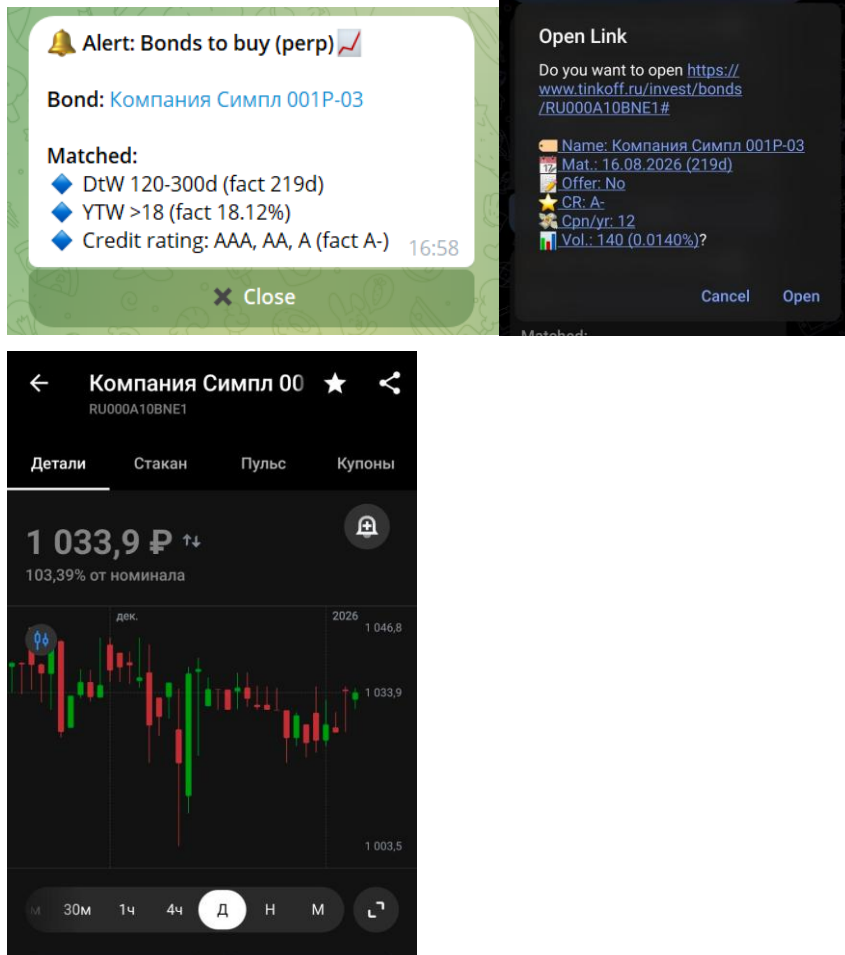
Close

7) Alert Notifications & Digest Messages

7.1 Instant Alert Notification (single bond)

When a bond triggers your alert, the bot sends a message like:

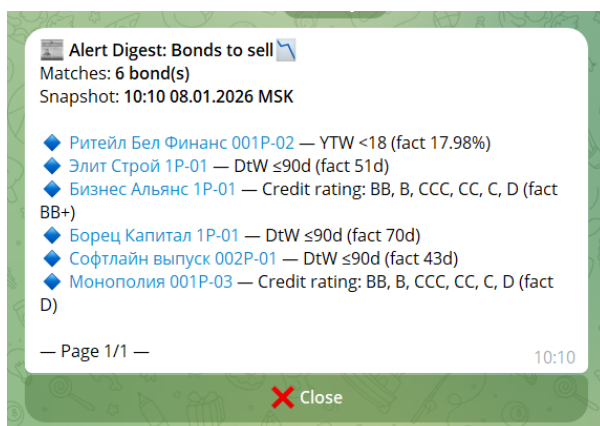
- Title: “🔔 **Alert:** ”
- Bond link line (which is also clickable and transforms to terminal or t-invest. The transition menu also contains important information on the bond)
- Matched conditions list (with values)
- Button: **✕ Close** (deletes the notification message)



7.2 Digest / Initial Check Pagination Controls

Digest and initial-check lists are paginated with:

- ◀ previous page
- X/Y page indicator (non-action)
- ▶ next page
- **✕ Close** (deletes digest message)



8) Troubleshooting & Tips

“Invalid expression”

- Ensure parentheses match.
- Ensure you did not leave dangling operators (e.g., “C1 AND” at the end).
- Ensure conditions exist before referencing them (don’t delete a condition token still used).

Too many notifications

- Use **Mute all** temporarily.
- Switch to **Daily digest** or **Weekly digest**.
- Use stronger filters or narrower scope.

Digest time not matching your local time

- Digest scheduling is based on **MSK (Europe/Moscow)** in code.
-

Watchlist (/watchlist, /wl) and Blacklist (/blacklist, /bl) functions

1) What Watchlist and Blacklist are (and why you'd use them)

Watchlist (👁👁)

Purpose: A personal “favorites” list of bond tickers you want to track.

What it's used for:

- **Alerts scope:** You can build alerts that monitor **only your Watchlist** (instead of the whole market or a custom set).
- **Screener view:** You can switch the screener to show **Watchlist-only** (so you don't see everything else).

Blacklist (🚫)

Purpose: A personal “exclude” list of bond tickers you don't want to see in the screener.

What it's used for:

- **Screener cleanup:** Removing unwanted bonds from the screener results (e.g., low liquidity, irrelevant issuers, duplicates).

What both lists “contain”

Both Watchlist and Blacklist store **bond tickers** (text codes like SU26238RMFS7).

2) How to open Watchlist / Blacklist panels

Watchlist

Send:

- /watchlist (or short alias /wl)

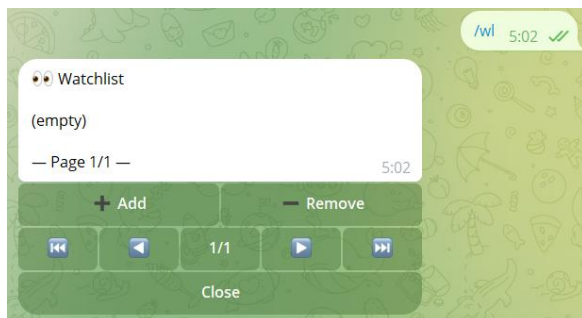
Blacklist

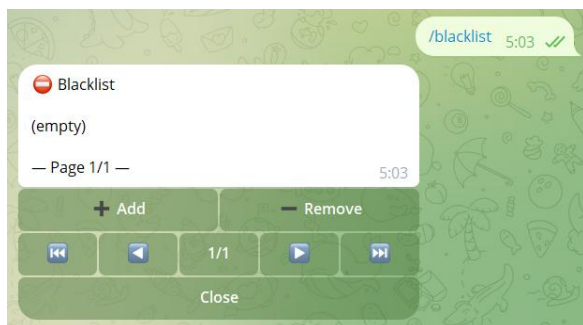
Send:

- /blacklist (or short alias /bl)

After you send the command, the bot posts a **panel message** with:

- A header (“👁👁 Watchlist” or “🚫 Blacklist”)
- A **paged list** (up to **23 items per page**)
- Buttons for navigation + Add/Remove/Close





3) How the list panel is laid out (what you're looking at)

Inside the panel, bonds are shown as:

- A **clickable ticker link** (opens the bond page) plus the bond's display name
- Grouped by sector (when sector data exists), with a section header like 🏢 SectorName: (N)
- A page footer like — Page X/Y —

If there's nothing to show, the panel displays (empty).

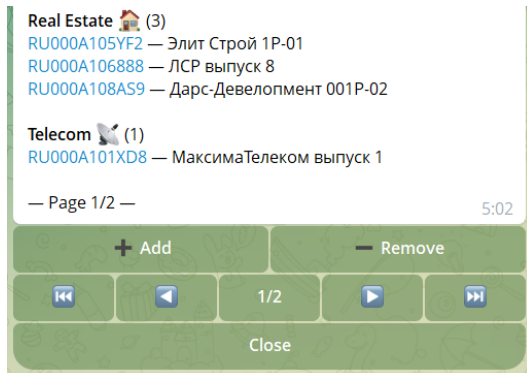


4) Every button explained

The Watchlist and Blacklist panels use the same button layout.

Navigation row

-  — go to first page
-  /  — go to previous page
- **X/Y** — page indicator
-  /  — go to next page
-  — go to last page



Action row

-  **Add** — start adding tickers to this list
-  **Remove** — start removing tickers from this list
-  **Close** — closes the panel and cleans up messages

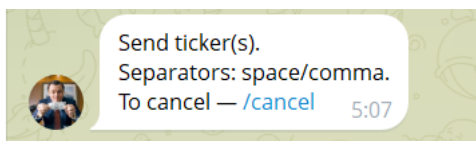
5) Adding bonds (step-by-step)

Works the same for Watchlist and Blacklist.

Step 1 — Click Add

The bot switches into “input mode” and sends a prompt:

“Send tickers. Separator: space or comma. To cancel — /cancel”



Step 2 — Type tickers and send

You can enter multiple tickers separated by:

- spaces, or
- commas

Examples:

- SU26238RMFS7 SU26239RMFS5

- SU26238RMFS7, SU26239RMFS5

Tickers are automatically uppercased internally.

Step 3 — Bot confirms and refreshes the panel

- If something changed. Bonds are added to the list.
- If everything you entered was already there: No changes.

Then the bot **updates the list panel** so you immediately see the result.

Also, the bot will delete:

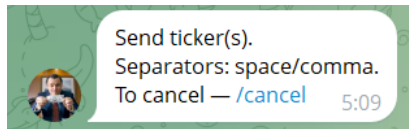
- your ticker message, and
- the prompt message
to keep the chat clean.

6) Removing bonds (step-by-step)

Works the same for Watchlist and Blacklist.

Step 1 — Click Remove

You'll get the same style prompt (space/comma-separated tickers; cancel via /cancel).



Step 2 — Type tickers you want removed and send

Same formatting rules as Add.

Step 3 — Bot confirms and refreshes the panel

- Removed: ... or No changes.
Then it refreshes the list panel.

7) Canceling Add/Remove mode

If you clicked **Add** or **Remove** but changed your mind:

Send:

- /cancel

The bot exits input mode and removes the input prompt.

8) Closing the Watchlist/Blacklist panel

Click:

-  Close

What it does:

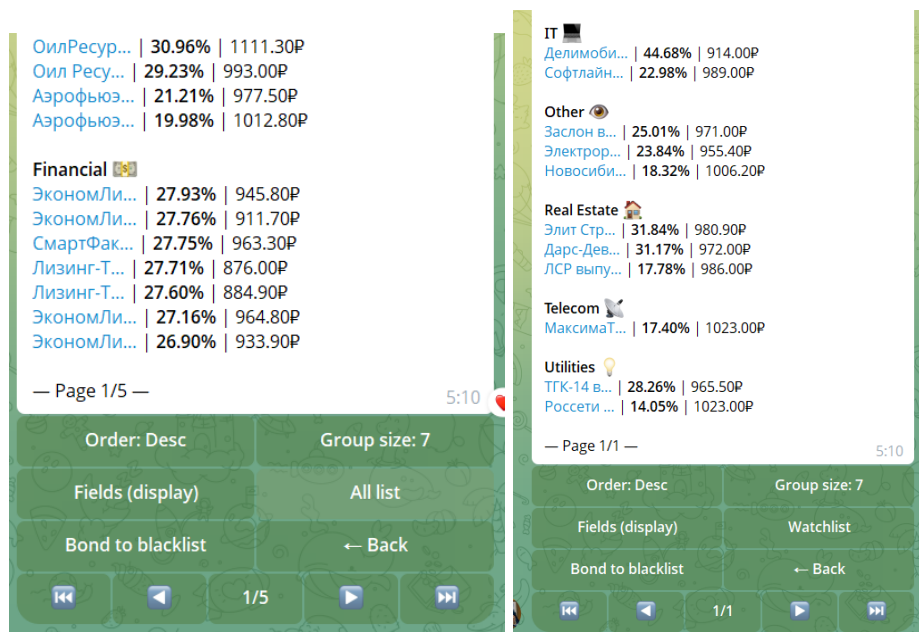
- Deletes the **panel message** and also removes the **original command message** (the /wl or /bl message), so your chat stays tidy.

9) Using Watchlist in the Screener (Watchlist-only mode)

Where it is

In the screener, open **Settings** and find the Watchlist toggle:

- It appears as “**Watchlist / All list**” (label changes depending on state).



What it does

- When switched to Watchlist mode, the screener is intended to show **only** bonds from your Watchlist.
- When switched back, the screener shows the full universe again (with your usual filters/sort/group).

10) Using Blacklist in the Screener (removing bonds)

Adding bonds to your Blacklist will exclude bonds from the panel of the screener.

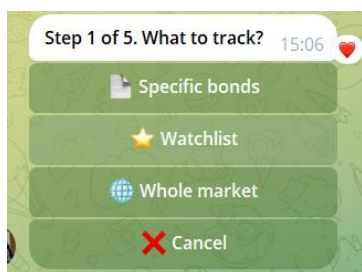
11) Watchlist vs Blacklist in Alerts (the important difference)

Watchlist is an Alerts scope

When creating an alert, the scope selection includes:

- ★ **Watchlist** (alongside “Specific bonds” and “Whole market”).

If you choose Watchlist scope, the alert evaluates **only bonds from your Watchlist**.



Blacklist does NOT affect alerts

Even when an alert scope is “Current screener”, the system explicitly ignores blacklist during alert evaluation. If some specific bonds should be excluded from the scope, button Remove bonds from the filter group should be used.

12) Practical notes & troubleshooting

1) “I added a ticker but I don’t see it in the list”

The list display is built from known tickers; the formatter filters through a ticker→UID mapping before it renders items.

So if a ticker is misspelled or not recognized, it may not appear as a clickable bond entry.

What to do:

- Re-check spelling
- Try searching/opening the bond elsewhere in the bot first, then re-add

2) “Why did my message disappear after I entered tickers?”

That’s expected: the bot will delete your input message and the prompt to keep the chat clean.

3) “How do I undo a mistake quickly?”

- Use **Remove** in the list panel, or
- open /bl and remove from Blacklist, or
- open /wl and remove from Watchlist

4) “How many items per page?”

Lists render with a page cap of **23 bonds per page**.
